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Weather Prediction Model using Savitzky-Golay and Kalman Filters

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Abstract

Daily Weather forecasting is essential in multiple areas like agriculture, energy supply, transportation and so on. It is a predictive challenge that has mainly relied on model based methods. We explore new techniques for forecasting weather with a data intensive approach. This proposed work studies, the power of making predictions via various deep learning approaches-a hybrid approach that combines a Long Short-term Memory Recurrent Neural network with Genetic Algorithm and a Statistical approach that uses the ARIMA (Auto Regressive Integrated Moving Average) model and an Estimation approach for the weather parameters taken into consideration with the usage of Filters. We show how the base model can be enhanced with the application of these approaches keeping in mind the dependencies between the parameters. Evaluation of these methods is done by experiments on real world meteorological data.

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Keywords: Weather; Prediction; Kalman filter; Savitzky-Golay; ARIMA;

1. INTRODUCTION

Weather forecasting is the utilization of science and technology to predict the conditions of the atmosphere for a given location and time. Weather forecasts are useful in various ways. Temperature and Precipitation forecasts are vital to agriculture, and hence, the traders who operate in these commodity markets. A time series is a collection of observations (data points) recorded over a period of time. Mostly, the data points are recorded at equally spaced time intervals. Weather forecast is performed by collecting observations of the current state of atmosphere and using knowledge of atmospheric processes to determine the state of atmosphere at a future point in time. Modern Weather forecasting primarily uses numerical models to make predictions. In Numerical weather prediction, forecast models (mathematical models) are applied to current observations of the weather data to make predictions. Currently the most effective way of forecasting the weather, the Numerical Weather Prediction method is not enough. The equations used in the models to simulate the atmosphere are chaotic in nature and impossible to solve without any errors. Several steps to predict the temperature are data collection (atmospheric pressure, temperature and humidity), data assimilation and analysis.

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1.1 Back Propagation Algorithm

Backpropagation is a technique utilized as a part of simulated neural systems to figure a slope that is required in the estimation of the weights to be utilized as a part of the system.[2] It is generally used to prepare profound neural systems, a term used to clarify neural systems with in excess of one shrouded layer.

1.2 Long Short-Term Memory Networks

Long Short Term Memory Networks typically just called “LSTMs” are an exceptional sort of RNN, fit for adapting long haul dependencies. Long term conditions are the point at which the RNN requires more data in the setting to anticipate the following information instead of just requiring the quick forerunner. LSTMs are expressly intended to stay away from the long haul reliance issue.

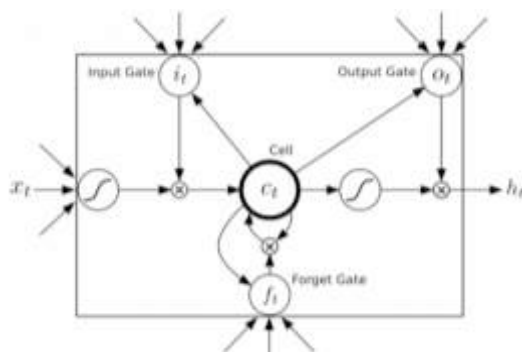


Fig 1. LSTM Architecture [Source:[17]]

1.3 Filters

Filters, majorly used in signal processing can be used in weather estimation. Known to have produced precise results previously, the two major-filters adopted in the proposed work are:

- Kalman Filter- It is an algorithm which utilizes a progression of estimations observed after some time and creates appraisals of the factors that have a tendency to be more exact than those when taken as a solitary estimation.
- Savitzky Golay Filter- It is a digital filter which can be connected to an arrangement of data points to smooth the data.

2. RELATED WORK

Many works were done related to prediction of weather. Santhosh Baboo et al [1] proposed an application of back propagation neural network for predicting the temperature based on the training set provided to the neural network. The algorithm improved convergence and damps the oscillations. This method proved to be a simplified conjugate gradient method. Yan Pei, et al [10] proposed two statistical ideas - fusion of uncertain estimates and unbiased linear estimators for correlated variables – which provides the conceptual foundation for Kalman filtering. By combining these ideas, standard results on Kalman filtering in linear systems was derived. Mohsen Hayati, et al, [7] used MLP(Multi-Layer Perceptron) network model used to predict short term temperature forecasting for Kermanshah City, in Iran, showed that MLP network has a good performance with minimum forecasting error. Shaminder Singh et al [2] proposed hybrid back propagation model using genetic algorithm to predict the weather. Y.Radhika et. al[5] showed the application of Support Vector Machine for weather prediction. Meng, XueMei[9] used improved genetic algorithm for weather prediction.

PROPOSED METHOD

Weather forecasting is an important problem that affects the everyday life of people. Forecasts, if made at the right time, can provide tremendous value in areas like agriculture, disaster detection etc. It can save lives and prevent destruction to property. It helps the general public be prepared on an everyday basis. Modern weather forecasting is performed using numerical models. With the advent of various learning techniques, which can generate rules and patterns from the observed non-linear data and train based on it, these learning techniques can be used to forecast weather data. The effectiveness of these approaches is studied and evaluated based on live data collected from the sensors.

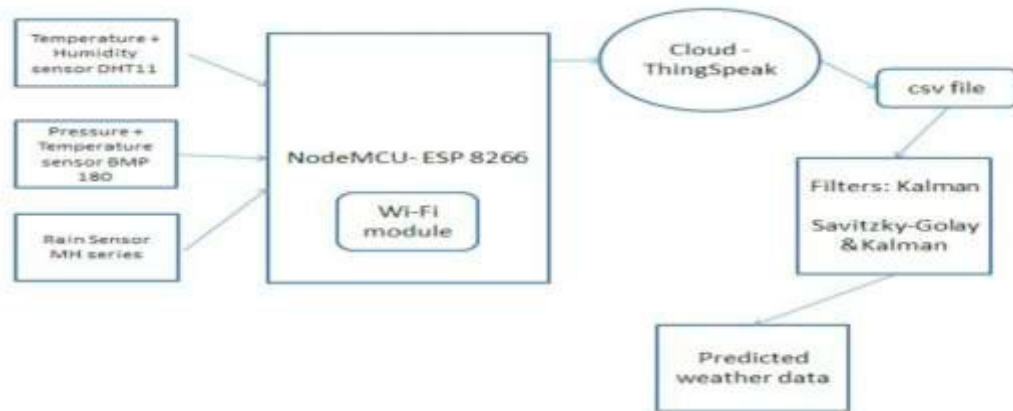


Fig 2. Framework of proposed system

A new combination- Savitzky Golay-Kalman filter has been used in this proposed work, wherein Savitzky Golay filter will be used to smoothen the data, then the data would be fed into the Kalman filter for prediction purposes.

3.1 Phases of Kalman Filter

It is a two-step process:

Prediction Step-The filter provides the estimates of the current state of the variable in consideration along with its uncertainties.

Update Step-In the update step, the estimates are up-dated using a weighted average once the outcome of the next measurement is observed.

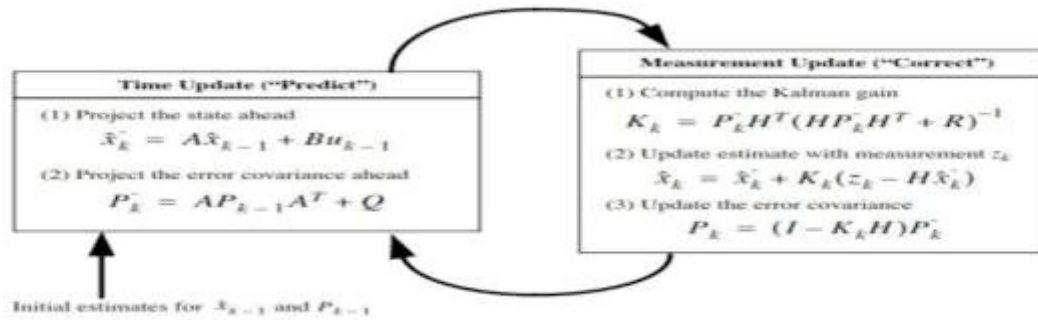


Fig 3. Kalman Filter Algorithm [Source: [15]]

3.2 Phases of Kalman-Savitzky Golay Filter in the proposed work

Savitzky-Golay filter is based on local least square polynomial approximation. In this filter, a polynomial is fit into a set of input samples and the resulting polynomial at a single point is evaluated within the approximation interval.

The combination of Kalman-Savitzky Golay filter was employed in the proposed work. Savitzky Golay filter was used to smoothen the data and the smoothened data was fed into Kalman Filter (eq-1) and predictions were observed.

$$(y_k) = \frac{\sum_{i=-n}^n A_i y_{k+1}}{\sum_{i=-n}^n A_i} \quad (1)$$

4.RESULT AND ANALYSIS

The proposed technique is tested with NodeMCU platform and the results have been compared with historic data and the graph is shown in fig-4.

4.1 NodeMCU

NodeMCU is an open source IoT platform. It includes firmware which runs on the ESP8266 Wi-Fi SoC and hardware which is based on the ESP-12 module.

Features:

- Interactive
- Programmable Low cost
- Simple

4.2 Auto Regressive Integrated Moving Average (ARIMA)

A measurable summed up model of the ARMA model, is a model which fit into time series information and it can be utilized for better comprehend information and anticipate future focuses in the series.

4.3 Dataset

Real-Time Data- Sensors were used to obtain real-time data. Two sensors-Temperature/Humidity Sensor and Pressure sensor were deployed to obtain the data. In the dataset, values for temperature, humidity, pressure and dew point were obtained.

Historic Data-The historic dataset adopted in the scenario consisted of about 14 attributes among which temperature, humidity, dew point and pressure were taken into consideration. Historic data has been taken from Kaggle website[16]. Attributes of the historic data are as follows:

['Date Time', 'p(mbar)', 'T(degC)', 'Tpot(K)', 'Tdew(degC)', 'rh(%)', 'VPmax(mbar)', 'VPact(mbar)', 'VPdef(mbar)', 'sh(g/kg)', 'H2OC(nmol/mol)', 'rho(g/m³)', 'wv(m/s)', 'max. wv(m/s)', 'wd(deg)']

4.4 Analysis

The proposed work lays emphasis on the results produced by the usage of filters namely, Kalman and Savitzky-Golay [14, 15]. In order to compare, the data sets were fit into ARIMA model. The mean squared error (MSE) is evaluated as the average of the squared forecast error values. The Root Mean Square Error (RMSE) helps in transforming the squared error into the original units of predictions by taking the square root of the squared score.

The data sets were analyzed using the four models namely:-

Neural Networks [1, 7, 4, 6],

Genetic Algorithm and LSTM [9],

ARIMA Model [11] and,

Filters-Kalman and Savitzky Golay and the following results were recorded.

The tables below, shows the statistics of the results that were observed:-

Table 1. RMSE values for the Historic Data for various models

Name of Model	Root Mean Squared Error (RMSE) - Historic Data
Neural Networks	0.2181
Hybrid of LSTM and Genetic Algorithm	1.4359906
ARIMA Model	0.2810693860
Kalman Filter	0.297613775
Kalman-Savitzky-Golay Filter	0.267852398

Amongst the four methods used, it is observed from the table that on usage of Neural Networks [1, 3], the RMSE is 0.2181 which is very less compared to the other methods. But, one needs to keep in mind the time complexity when analyzing the performance of various models with respect to the data set. The usage of filters with the data set produced RMSE of 0.2976(Kalman Filter) and 0.2678(SG-Kalman) suggesting that the usage of filters produced optimized results.

Table 2. Comparison of RMSE values for the Historic and Real time Data

Name of Model	Root Mean Squared Error (RMSE) Historic Data	Root Mean Squared Error (RMSE) Real time Data
ARIMA Model	0.2810693860	0.29287528
Kalman Filter	0.297613775	0.1480668
Kalman-Savitzky-Golay Filter	0.267852398	0.11904551

Keeping in mind the time complexity associated on the implementation of neural network model, for comparison purposes, only the other three models were used. Amongst the three methods used for the comparison of both the data sets, it is observed from the table that on usage of ARIMA Model, the RMSE is 0.2810 for historic data and 0.2928 for real-time data. The usage of filters with the data sets produced RMSE of 0.2976 for historic data and 0.1408 with respect to Kalman Filter and 0.2678 for historic data and 0.1990 for real-time data with respect to Savitzky Golay -Kalman suggesting that the usage of filters produced optimized results. The results of various methods obtained have been visualized in the form of a graph as shown below.

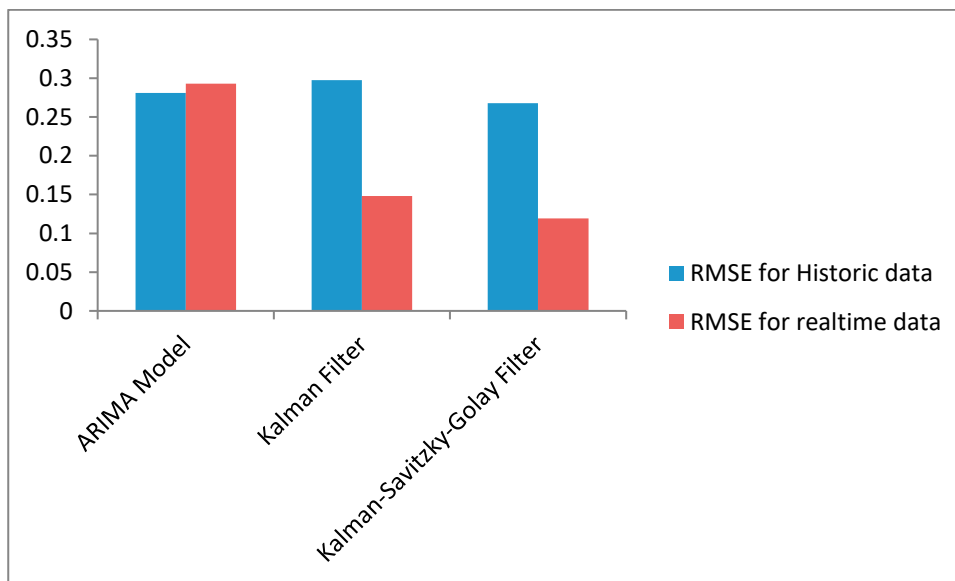


Fig 4. Comparison of RMSE and MSE for various models

In the figure, the brown colour represents the “Historic Data” and the blue line represents the “Real-Time Data”. NodeMCU was the main component used in the proposed work. DHT11 sensor used in the proposed work proved to be the best choice, since it provided values instantly for various attributes. The components used were found to be a cost-effective solution for the weather prediction.

3. CONCLUSION

The paper has explored alternative models for weather forecasting. In this paper, as a first step towards forecasting, neural networks was used to predict with low error. But the limitation of neural networks was the time it took to run on CPU.

To reduce the time complexity, the adoption of the statistical ARIMA model provided an insight on how the prediction of parameters needs to be done (including parameter dependencies) and results were produced with low error as compared to the first model. With the aim to reduce the complexity even further, Filters (such as Kalman and Kalman-Savitzky-Golay) was adopted for forecasting. It was observed that the usage of filters enabled fast processing and improved forecasts of the parameters that were taken into consideration compared to the first two models. Hence, the usage of filters is a recommended choice for the future estimation of scientific weather data.

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